

VULNERABILITY RESEARCH

ASSIGNMENTS

SPECIALISATION

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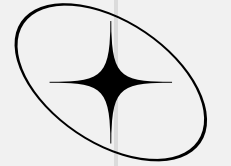
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COMMON TERMINOLOGY

The below are common terms that will be used through this course



GOOGLE SLIDES

CVSS

THE COMMON VULNERABILITY SCORING SYSTEM (CVSS) IS A STANDARDIZED FRAMEWORK FOR RATING THE SEVERITY OF SECURITY VULNERABILITIES. IT HELPS ORGANIZATIONS PRIORITIZE THEIR RESPONSES TO VULNERABILITIES BASED ON THEIR IMPACT AND EXPLOITABILITY.

BINARY

A BINARY FILE IS A COMPUTER FILE THAT CONTAINS ENCODED DATA THAT IS NOT HUMAN-READABLE. IN THE CONTEXT OF SOFTWARE, IT USUALLY REFERS TO EXECUTABLE FILES COMPILED FROM SOURCE CODE.

PWN

THIS IS A TERM DERIVED FROM THE WORD "OWN," USED TO DESCRIBE THE ACT OF COMPROMISING OR CONTROLLING ANOTHER COMPUTER, NETWORK, OR SERVICE.

PATCH DIFFING

THE PROCESS OF COMPARING DIFFERENT VERSIONS OF SOFTWARE PATCHES TO IDENTIFY CHANGES. THIS TECHNIQUE CAN BE USED TO UNDERSTAND WHAT VULNERABILITIES HAVE BEEN FIXED AND SOMETIMES TO IDENTIFY NEW VULNERABILITIES INTRODUCED BY THE PATCH.

MALWARE

MALICIOUS SOFTWARE DESIGNED TO HARM, EXPLOIT, OR OTHERWISE COMPROMISE THE OPERATION OF A COMPUTER SYSTEM. COMMON TYPES INCLUDE VIRUSES, WORMS, TROJANS, RANSOMWARE, AND SPYWARE.

PATCH TESTING

PATCH TESTING IS WHERE, IT AND SECURITY TEAMS AIM TO DETECT AND FIX PROBLEMS (IN PATCHES) BEFORE THEY IMPACT THE ENTIRE NETWORK

EXPLOIT

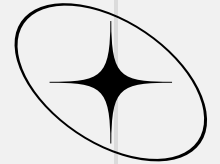
A PIECE OF SOFTWARE, A CHUNK OF DATA, OR A SEQUENCE OF COMMANDS THAT TAKES ADVANTAGE OF A BUG OR VULNERABILITY TO CAUSE UNINTENDED OR UNANTICIPATED BEHAVIOR IN SOFTWARE OR HARDWARE.

FUZZING

AN AUTOMATED SOFTWARE TESTING TECHNIQUE THAT INVOLVES PROVIDING INVALID, UNEXPECTED, OR RANDOM DATA INPUTS TO A PROGRAM TO FIND SECURITY VULNERABILITIES AND BUGS.

COMMON TERMINOLOGY

The below are common terms that will be used through this course



GOOGLE SLIDES)

SHELLCODE

A SMALL PIECE OF CODE USED AS THE PAYLOAD IN THE EXPLOITATION OF A SOFTWARE VULNERABILITY. SHELLCODE IS TYPICALLY USED TO LAUNCH A COMMAND SHELL ON THE TARGET MACHINE, GIVING THE ATTACKER CONTROL OVER IT.

SECURITY ADVISOR

A PROFESSIONAL WHO PROVIDES EXPERT ADVICE ON HOW TO PROTECT SYSTEMS, NETWORKS, AND DATA FROM CYBERSECURITY THREATS. THEY HELP ORGANIZATIONS IMPLEMENT SECURITY POLICIES, STRATEGIES, AND MEASURES.

0 - DAY

A ZERO-DAY VULNERABILITY IS A PREVIOUSLY UNKNOWN SECURITY FLAW THAT HACKERS CAN EXPLOIT BEFORE THE SOFTWARE DEVELOPER HAS RELEASED A FIX. IT IS CALLED "ZERO-DAY" BECAUSE DEVELOPERS HAVE ZERO DAYS TO ADDRESS THE ISSUE ONCE IT IS DISCOVERED.

N - DAY

AN N-DAY VULNERABILITY IS A KNOWN SECURITY FLAW FOR WHICH A PATCH HAS BEEN RELEASED, BUT NOT ALL SYSTEMS HAVE BEEN UPDATED TO FIX IT. HACKERS EXPLOIT THESE KNOWN VULNERABILITIES IN SYSTEMS THAT HAVE NOT YET BEEN PATCHED.

OS INTERNALS

REFERS TO THE INTERNAL COMPONENTS AND OPERATIONS OF AN OPERATING SYSTEM, INCLUDING KERNEL, DRIVERS, SYSTEM CALLS, AND OTHER LOW-LEVEL FUNCTIONALITIES.

VULNERABILITY

A FLAW OR WEAKNESS IN A SYSTEM'S DESIGN, IMPLEMENTATION, OR OPERATION THAT COULD BE EXPLOITED TO VIOLATE THE SYSTEM'S SECURITY POLICY.

MACHINE

IN THE CONTEXT OF CYBERSECURITY, "MACHINE" USUALLY REFERS TO A COMPUTER OR DEVICE THAT CAN BE TARGETED OR USED IN A SECURITY OPERATION.

CODE REVIEW

THE SYSTEMATIC EXAMINATION OF SOURCE CODE BY DEVELOPERS OR SECURITY EXPERTS TO FIND AND FIX SECURITY VULNERABILITIES, IMPROVE CODE QUALITY, AND ENSURE ADHERENCE TO CODING STANDARDS.